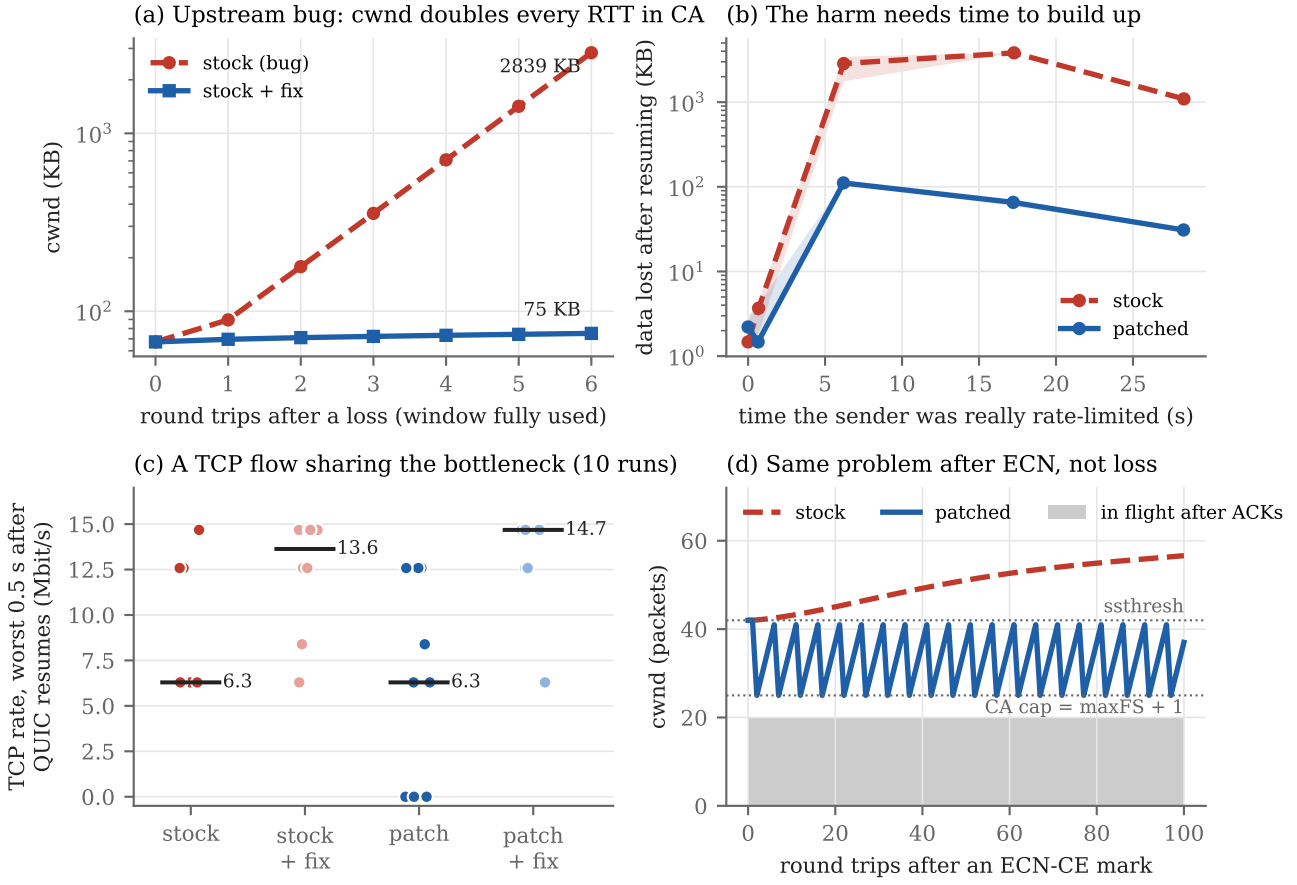
(a) Upstream bug: cwnd doubles every RTT in CA
stock (bug)
stock + fix
2839 KB
75 KB
cwnd (KB)
round trips after a loss (window fully used)
(b) The harm needs time to build up
data lost after resuming (KB)
stock
patched
time the sender was really rate-limited (s)
(c) A TCP flow sharing the bottleneck (10 runs)
TCP rate, worst 0.5 s after QUIC resumes (Mbit/s)
13.6
14.7
6.3
6.3
stock
stock + fix
patch
patch + fix
(d) Same problem after ECN, not loss
stock
patched
in flight after ACKs
ssthresh
CA cap = maxFS + 1
cwnd (packets)
round trips after an ECN-CE mark